

K1426 — primary-source pin on the D_{IV^5} Bergman exponent (settles the $7/2$ -vs- $5/2$ -vs-5 tangle, backs K1424, sharpens Lyra’s weight fork). ★ Authoritative FK/Bergman fact: for an irreducible bounded symmetric domain the Bergman kernel is $K(z,w) = h(z,w)^{-\gamma}$, where h is the generic norm and γ is the genus, $\gamma = 2 + a(r-1) + b$. For type IV_n (the Lie ball) the genus is $\gamma = n$, so for D_{IV^5} : $\gamma = 5 = n_C$ ($a=3=N_C$, $b=0$, $r=\text{rank}=2$, $\gamma = 2+3\cdot 1+0 = 5$). ★ The scalar Bergman exponent is the genus $= 5 = n_C$. $g = 7$ does NOT appear in it — $g=7$ is the signature total $p+q$, exactly as K1424 said. Paper118’s “ $-g/\text{rank} = -7/2$ ” mislabels the signature g as the genus; confirmed wrong from the primary source. ★ The 5 -vs- $5/2$ difference is a *base-normalization convention* ($\gamma=5$ on the generic norm h of degree $r=2$; the “ $5/2$ ” appears only in a rank-halved base), NOT a disagreement about the invariant — the genus is unambiguously $5=n_C$, never 7. ★ SHARPENED FORK for Lyra’s fermionic $s=7/2$: it is correct only if BOTH (a) her G -base is the rank-halved one so the *scalar* weight reads $5/2 = n_C/\text{rank}$, AND (b) the spin lift adds exactly $+1$ ($5/2 \rightarrow 7/2$, the metaplectic/half-form shift). Both must be *derived*; if her base is the generic-norm convention (scalar $= 5$), then $7/2$ is wrong. Cal cold-reads.

Keeper (2026-08-12, ~22:15. Casey reading the QM paper tonight; I took the weight thread because it’s my flag and I got burned citing that exponent from memory this afternoon. Pinned it to the FK genus formula from the literature so there’s one authoritative answer instead of three floating in the corpus. Cal cold-reads. Nothing pushed.)

★ The authoritative fact (primary-source, target-innocent)

For an irreducible bounded symmetric domain Ω , the Bergman kernel is $K(z,w) = h(z,w)^{-\gamma}$, h = the unique generic norm (degree = rank r), γ = the genus $= 2 + a(r-1) + b$, with a, b the FK root multiplicities.

For type IV_n (the Lie ball, complex dimension n): tube type, $r = 2$, $a = n-2$, $b = 0$ □ $\gamma = 2 + (n-2)\cdot 1 + 0 = n$. So: D_{IV^5} : genus $\gamma = 5 = n_C$ ($a = 3 = N_C$, $b = 0$, $r = 2$). Matches Cal’s independently-ratified FK-roots value (genus $p = (r-1)a + 2 = 5$).

□ the scalar Bergman exponent of D_{IV^5} is the genus $= 5 = n_C$. (Sources: the FK genus/generic-norm formula $K = h^{-\gamma}$; standard for bounded symmetric domains including Lie balls — see the generic-norm definition $N = (V\cdot K)^{-1/\gamma}$ in the literature below.)

★ What this settles

- $g = 7$ does NOT appear in the scalar Bergman exponent. The exponent is genus-governed ($= 5 = n_C$); $g = 7$ is the *signature total* $p + q$, a different invariant. This is primary-source confirmation of K1424 — Paper118’s “ $K_B = \det(1-z\bar{w}^T)^{-g/\text{rank}} = D^{-7/2}$ ” uses $g=7$ as if it were the genus, which is precisely the mislabel my State-of-Spine audit corrected. Paper118 lines 41–43 are wrong on the exponent; the relabel stands (Grace’s $7/2$ sweep).
- The 5 -vs- $5/2$ is a convention, not a conflict. $\gamma = 5$ is the exponent on the *generic norm* h (degree $r = 2$). A “ $5/2$ ” appears only if the base is rank-halved ($h = D^2$, exponent γ/rank per D -factor) or written per-Jordan-trace. Either way the genus is $5 = n_C$, never 7. The corpus should state the *base* alongside the exponent

and stop trading bare fractions (this is the same “pin the base, not just the number” discipline that caused the tangle).

★ Sharpened fork for Lyra’s fermionic weight (the F947 / K1425 open item)

Lyra’s exact projector uses $P = S(B/G) \cdot G^{-s}$, $s = 7/2$. Given the pin above, $7/2$ is correct only if BOTH conditions hold, and both must be derived, not assumed: 1. Base condition: her G is the rank-halved base, so the *scalar* weight reads $5/2 = n_C / \text{rank} = \gamma / \text{rank}$ (not the generic-norm base, which would read 5). She must state which base G is. 2. Spin-shift condition: the spin lift $S(B/G)$ contributes exactly $+1$ to the weight (the metaplectic / half-form / canonical-bundle twist), giving $5/2 + 1 = 7/2$. - If both hold and are derived, then $7/2$ is the *derived* fermionic weight — the “why $g = 7$ lives in the fermion kernel” K1424 asked for, and a genuine result: $g = 7 = 2 \cdot (5/2) + 2 \cdot (1/2) \dots = (\text{scalar } 5/2) + (\text{spin } +1)$, i.e., the spinor weight, not the genus. Worth banking *as a derivation*. - If her base is the generic-norm one (scalar weight 5), then $5 + \text{spin-shift} \neq 7/2$ and the weight is wrong. - The one check that decides it (Lyra + Cal, per K1424): does the reproducing/idempotency condition ($P \cdot P = P$ under the invariant measure) *force* $s = 7/2$? Hermitian-symmetry alone (what F947 verified) does *not* pin s . Pin the base, derive the spin shift, then s is forced — or it isn’t.

Ratio-of-invariants note (for the ledger)

The clean, base-free statement to carry forward: scalar Bergman weight = genus = $n_C = 5$; fermionic (spinor) weight = scalar + spin-lift shift; $g = 7$ enters only as the signature, never as the genus. Any corpus formula that writes a Bergman exponent as “ g/rank ” is carrying the retired mislabel and should be relabeled to the genus.

Route

1. Lyra — for the weight fork: (a) state which base G is (generic-norm $\rightarrow 5$, or rank-halved $\rightarrow 5/2$); (b) derive the spin-lift shift (is it exactly $+1$?); (c) confirm the reproducing condition *forces* s , don’t rest on Hermitian-symmetry. Then $7/2$ is derived-or-refuted on solid ground.
2. Grace — the Paper118 $7/2$ relabel (K1424) now has primary-source backing: relabel to genus = $n_C = 5$ (state base), not “ g/rank .”
3. Cal — weight-fork cold-read (K1425 §4) now has a concrete rubric: $7/2$ requires rank-halved base ($5/2$) + a derived $+1$ spin shift; reject if either is assumed.
4. Keeper — pinned the exponent to the primary source; K1424 confirmed; fork sharpened to two derivable conditions. Recused from Lyra’s derivation; will adjudicate against this ground truth when it lands.

Sources: FK genus / generic-norm & Bergman kernel $K = h^{-\gamma}$ (weighted Bergman kernels on bounded symmetric domains); generic norm $N = (V \cdot K)^{-1/\gamma}$, Schatten-class Bergman/Szegő on bounded symmetric domains, arXiv:2009.09613.

— Keeper, K1426, 2026-08-12. PRIMARY-SOURCE PIN on the D_{IV}^5 Bergman exponent. FK fact: $K(z,w) = h(z,w)^{-\gamma}$, $h = \text{generic norm (degree=rank } r)$, $\gamma = \text{genus} = 2 + a(r-1) + b$; type $IV_n \rightarrow \gamma = n$; $D_{IV}^5 \rightarrow \gamma = 5 = n_C$ ($a=3=N_C, b=0, r=2$), matches Cal’s ratified FK-roots. [?] SCALAR BERGMAN EXPONENT = GENUS = $5 = n_C$; $g=7$ does NOT appear ($g=7$ is signature total $p+q$). CONFIRMS K1424: Paper118 “ $-g/\text{rank} = -7/2$ ” mislabels g as genus, WRONG from primary source; relabel stands. $5\text{-vs-}5/2 = \text{base-normalization convention}$ ($\gamma=5$ on generic norm degree-2; $5/2$ only in rank-halved base), NOT a conflict — genus unambiguously $5 = n_C$ never 7; corpus must state BASE not trade bare fractions. SHARPENED FORK (Lyra F947 $s=7/2$): correct ONLY IF (a) G is rank-halved base so scalar reads $5/2 = n_C / \text{rank}$ AND (b) spin lift adds exactly $+1$ ($5/2 + 1 = 7/2$), both DERIVED; if base is generic-norm (scalar=5) then $7/2$ wrong. Decider (Lyra+Cal, K1424): does reproducing/idempotency FORCE s ? Hermitian-sym alone doesn’t pin s . Ledger statement: scalar weight=genus= $n_C=5$; fermionic=scalar+spin-shift; $g=7$ enters only as signature never genus; any “ g/rank ” Bergman exponent carries the retired mislabel→relabel to genus. Route: Lyra state base+derive spin-shift+confirm reproducing forces s ; Grace relabel Paper118 to genus= $n_C=5$ (primary-source-backed); Cal weight-fork rubric (rank-halved $5/2$ + derived $+1$, reject if assumed); Keeper pinned exponent, K1424 confirmed, fork sharpened, recused. Cal cold-reads. Nothing pushed.